Solution Description

1 Datasets Used

Logic Dataset (Merged & Augmented):

- **Name/Source:** EXACT Logic Merged & Augmented Set (`logic_merged_valid_augmented.json`). A merge of EXACT 2026 Official Logic-Based Educational Queries (`logic_based.json`, 411 records) and the FOLIO dataset, augmented via quantifier variable canonicalization, entity anonymization, multi-premise permutation, hard negative generation, and synthetic multi-hop logic reasoning problems.
- **Samples:** 1,789 total samples (1,715 train, 74 validation).
- **Sample Entry:**
  Input (NL): "If a student is enrolled in a science program and has passed Chemistry 101, they can enroll in Organic Chemistry." → *Question:* "Based on the premises, can we conclude that Minh can enroll in Organic Chemistry?"
  Output (FOL): $\forall x(\text{science\_program}(x) \land \text{passed}(x, \text{chem101}) \rightarrow \text{can\_enroll}(x, \text{organicchem}))$
  Output (Ans & Explanation): "Yes" → "Minh is enrolled in a science program and passed Chemistry 101, so the premise implies he can enroll..."

Physics Dataset (Augmented):

- **Name/Source:** EXACT Physics Augmented Dataset (`physics_augmented.json`). Preprocessed from EXACT 2026 Official Physics problems and enhanced with structured synthetic reasoning chains.
- **Samples:** 1,371 total records.
- **Sample Entry:**
  Input (Question): "Two test charges q_1 and q_2 ($q_1 = 4q_2$)..."
  Output (JSON):
  `{"thought": "...", "physics_analysis": [...], "algebraic_reasoning": [...], "python_code": "..."}`

2 Approach and Method

Our system uses a standard orchestration layer routing incoming queries to specialized domain pipelines:

Logical Reasoning Pipeline (`type1`): (1) *Glossary-Constrained Translation:* Compiles NL premises and questions into FOL by pre-building a unified JSON glossary of predicates and constants. (2) *FOL Normalization & Repair Loop:* Standardizes quantifiers and connectives, using an LLM-based repair agent to resolve syntax/casing errors. (3) *Z3 SMT Solver:* Mathematically verifies logical entailment. (4) *Neurosymbolic Proof-Guided CoT:* Extracts the minimal Z3 unsat core (if entailed) to guide the LLM in generating a clear, hallucination-free explanation.

Physics Solving Pipeline (`type2`): (1) *Input Preprocessing:* Sanitizes raw input, performs canonical variable extraction, and executes SI unit normalization to ensure all physical quantities are in consistent base units before policy application. (2) *Domain Classification:* Routes physics queries into domain categories (e.g., `electrostatic_force`, `circuit_power`). (3) *Reasoning Policy Assembly:* Dynamically retrieves domain-specific reasoning policies and few-shots based on the domain classification. (4) *LLM Code-Gen Solver:* Generates step-by-step reasoning and a Python calculation script. (5) *Sandboxed Execution:* Runs the code for precise unit/circuit calculations. (6) *Postprocessing:* Applies rounding rules and converts result scales to engineering prefixes (e.g., 1×10^{-3} N to 1 mN).

3 Model Size Calculation

Our pipeline is powered by a single base model with dynamic task-specific adapters loaded and swapped on demand:

Model Component	Total Parameters	Active Parameters
Base LLM (`Qwen/Qwen3-8B`)	8.0B	8.0B
`fol_router` Adapter (Rank-64 LoRA)	$\approx$ 0.35B	$\approx$ 0.35B (When active)
`physics` Adapter (Rank-64 LoRA)	$\approx$ 0.35B	$\approx$ 0.35B (When active)
Runtime Peak Footprint	**8.7B (Loaded)**	**8.35B (Class-compliant)**

Verification: Both Rank-64 LoRA adapters target all projection modules. The API server loads the base model and both adapters into memory, utilizing PEFT's dynamic adapter swapping (`set_adapter`). Since only one adapter is active during any single inference forward pass, the active parameters are limited to $8.0\text{B} + 0.35\text{B} = 8.35\text{B}$, fully complying with the 8B-class limit.